CS 131 Project Proposal

1. Problem Statement & Originality

Modern sports analytics relies on spatial player tracking to optimize tactical spacing and team structures. While professional stadiums use expensive, multi-camera rigs, amateur teams are restricted to single-camera broadcast or mobile footage. Reconstructing a 2D top-down map from this source is difficult due to three constraints: lens distortion from wide-angle views, sideline clutter (coaches, substitutes, and referees), and camera ego-motion, which constantly shifts the pixel coordinate system and pushes critical boundary lines completely out of the frame. The originality of this project lies in bypassing full 3D multi-camera environment reconstruction. Instead, it utilizes single-view planar homography constraints and algebraic line intersection to map active player coordinates onto a fixed, top-down metric field.

2. Proposed Methodology

- **Stage I: Perception & Roster Filtering:** A deep convolutional network (YOLO Architecture) detects humans in the frame. To eliminate sideline spectators and enforce strict roster limits detections are ranked dynamically by confidence scores and spatial bounding box boundaries.
- **Stage II: Geometric Lens Undistortion (if applicable):** To fix barrel lens distortion where field lines curve near corners, we apply multi-parameter radial and tangential correction factors. This mathematically rectifies the discrete anchor coordinates (the bottom-center pixel of player bounding boxes) into an ideal coordinate system where lines are linear.
- **Stage III: Structural Extrapolation:** For frames where camera panning pushes sidelines and endlines out of bounds, we deploy a line-fitting algorithm on the remaining visible boundary markers. By computing the intersection of these lines using matrix determinants, we mathematically synthesize the coordinates of theoretical, out-of-frame field corners.
- **Stage IV: Homography Transformation:** Using the four reference corners, we compute a matrix which maps the camera coordinates into a top-down tactical metric canvas for visualization.

CS 131 Relevance: This pipeline directly implements foundational course concepts including planar homography, camera matrix models, lens distortion coefficients, and projective geometry.

3. Feasibility & 4-Week Timeline

- **Week 1: Perception & Detection Optimization**
  Objectives: Extract video frames; initialize the detection network to isolate player coordinates.
- **Week 2: Lens Calibration Framework**
  Objectives: Implement lens correction equations to flatten field boundaries mathematically.
- **Week 3: Line Intersection & Homography Loop**
  Objectives: Develop line-fitting and intersection logic to compute theoretical, out-of-frame corners; calculate the dynamic homography translation matrix.
- **Week 4: Mapping Canvas & Validation**
  Objectives: Generate the 2D canvas plotting wrapper; validate coordinates against known field dimensions; compile the final project report.